Decoding Parameter and Sampling Analysis

Impact of Temperature, Top-K and Top-P on Determinism, Hallucination and Output Structure in Llama 3 and Qwen 2.5:3b

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Abstract

Large language models generate text by repeatedly sampling the next token from a probability distribution. Decoding parameters such as Temperature, Top-K and Top-P reshape this distribution and therefore control how random, accurate and well-formatted the output is. This report measures the effect of these three parameters on two locally hosted models, Llama 3 and Qwen 2.5:3b, run through Ollama. Three behaviours are evaluated: **determinism** (similarity of repeated answers), **hallucination** (factual questions and fake-entity trap questions) and **structure** (strict JSON and numbered-list formatting). One parameter is varied at a time while the other two stay at a baseline (Temperature 0.7, Top-K 40, Top-P 0.9), giving 9 settings per model and 972 generated answers in total.

The main findings are: (1) Top-K = 1 (greedy decoding) is the only setting that makes outputs fully identical, giving 100% similarity for both models, while every other setting produced three different answers in three runs; (2) Qwen 2.5:3b hallucinates far more than Llama 3 (about 12.5% to 20.8% versus 0% to 8.3% by the automatic metric), mainly because it invents a summary for a non-existent research paper in every setting; (3) Qwen follows strict formats perfectly (100%) while Llama 3 occasionally breaks strict JSON; and (4) the effect of Temperature and Top-P on most metrics is small compared to the sampling noise in this small experiment, so those trends are reported as indicative rather than conclusive.

1. Introduction

When a language model writes a word, it first computes a score for every token in its vocabulary and converts the scores to probabilities. The decoding strategy then decides which token to pick. Three common controls are:

- **Temperature** rescales the probabilities. Low values make the model prefer its most likely tokens; high values flatten the distribution and make unlikely tokens more probable.
- **Top-K** keeps only the K most probable tokens at each step. K = 1 always selects the single most probable token (greedy decoding).
- **Top-P (nucleus sampling)** keeps the smallest set of tokens whose probabilities add up to P, so the number of candidates adapts to how confident the model is.

The aim of this project is to measure, rather than assume, how these controls change the behaviour of two small open models, so that sensible parameter values can be chosen for different tasks such as factual question answering, creative writing and structured data generation.

1.1 Research questions

- How deterministic are repeated answers to the same prompt under different decoding settings?
- Do higher randomness settings increase hallucination, including for questions about things that do not exist?
- Do higher randomness settings reduce the ability to obey strict output formats?
- How do Llama 3 and Qwen 2.5:3b differ in these respects?

2. Methodology

2.1 Models and software

Component	Details
Models	Llama 3 (tag *llama3*) and Qwen 2.5:3b (tag *qwen2.5:3b*), served locally by Ollama
Interface	Ollama REST endpoint */api/generate* with *stream = false*
Application	LLM Sampling Lab, a Streamlit app (*app.py*) that runs the experiments, stores raw answers as CSV and draws the charts
Libraries	Python, Streamlit, Pandas, NumPy, scikit-learn (TF-IDF and cosine similarity), Plotly, Requests, Matplotlib and ReportLab (this report)
Answer length limit	num_predict = 250 tokens (determinism), 120 (hallucination), 220 (structure)

2.2 Experimental design

Each decoding parameter is swept over three values while the other two remain at the baseline. The baseline setting therefore appears once in every sweep, which also lets us observe the natural run-to-run noise.

Parameter swept	Values tested	Fixed at
Temperature	0.2, 0.7, 1.2	Top-K 40, Top-P 0.9
Top-K	1, 20, 40	Temperature 0.7, Top-P 0.9
Top-P	0.5, 0.9, 1.0	Temperature 0.7, Top-K 40

2.3 Tasks

Experiment	Prompts	Runs per setting	Answers per model
Determinism	One open-ended prompt: *Explain Artificial Intelligence in simple words.*	3	27
Hallucination	5 factual questions (capital of Australia, author of 1984, end of World War II, symbol of gold, general relativity) and 3 trap questions about non-existent things (a fake 2019 paper by Dr. Marlon Vexley, a 1987 Nobel Prize in Mathematics, and the population of Zorlandia in Norway)	3	216
Structure	A strict JSON object with keys name, definition and examples (exactly 3 strings); and a numbered list of exactly 5 items with no introduction or conclusion	3	54

2.4 Metrics

- **Determinism**: mean pairwise TF-IDF cosine similarity between the three answers (100% means identical wording), plus the number of unique answers. Diversity is 100 minus similarity.
- **Hallucination**: for a factual question the answer is correct if it contains the expected keyword; it counts as hallucinated if it is wrong and does not admit uncertainty. For a trap question the correct behaviour is to say the entity does not exist or is unknown; confidently answering is a hallucination. This is an automatic keyword heuristic.
- **Structure**: *strict pass* requires the whole output to be exactly the requested JSON object (or exactly 5 numbered lines and nothing else); *parseable* is the lenient version that allows extra text.

3. Expected Results

Before running the experiments the following behaviour was expected, based on how the three parameters work:

Metric	Expected effect of more randomness (higher T, K, P)
Determinism	Similarity falls as Temperature, Top-K and Top-P rise. Top-K = 1 should give identical answers regardless of Temperature.
Hallucination	Hallucination rate rises, because lower-probability (often wrong) tokens are sampled more often. The smaller model (Qwen 2.5:3b) should hallucinate more than Llama 3.
Structure	Strict-format pass rate falls, since a single stray token can break valid JSON or add extra text.

4. Results

4.0 Overall results per model

Table 0 gives the headline value of each metric, averaged over all nine settings of a model (these are the same numbers shown in the summary cards of the LLM Sampling Lab app). The determinism, hallucination and structure sections below break these averages down by parameter. Note that the 11.1% fully identical runs come entirely from the single Top-K = 1 setting (1 of 9 settings).

Metric	Llama 3	Qwen 2.5:3b
Determinism: text similarity (%)	65.9	53.6
Determinism: fully identical runs (%)	11.1	11.1
Determinism: diversity proxy (%)	34.1	46.4
Determinism: average words	192.7	134.1
Hallucination: accuracy (%)	97.7	85.2
Hallucination: rate (%)	2.3	14.8
Structure: strict format pass (%)	88.9	100.0
Structure: parseable (%)	88.9	100.0

Table 0. Overall averages over all settings.

4.1 Determinism

Figure 1 shows how similar the three repeated answers were. The most striking result is Top-K = 1: both models returned word-for-word identical answers in every run, giving 100% similarity and exactly one unique output. For every other setting, all three runs were different (3 unique outputs, 0% fully identical answers), even at Temperature 0.2, because Top-K 40 and Top-P 0.9 still allow sampling.

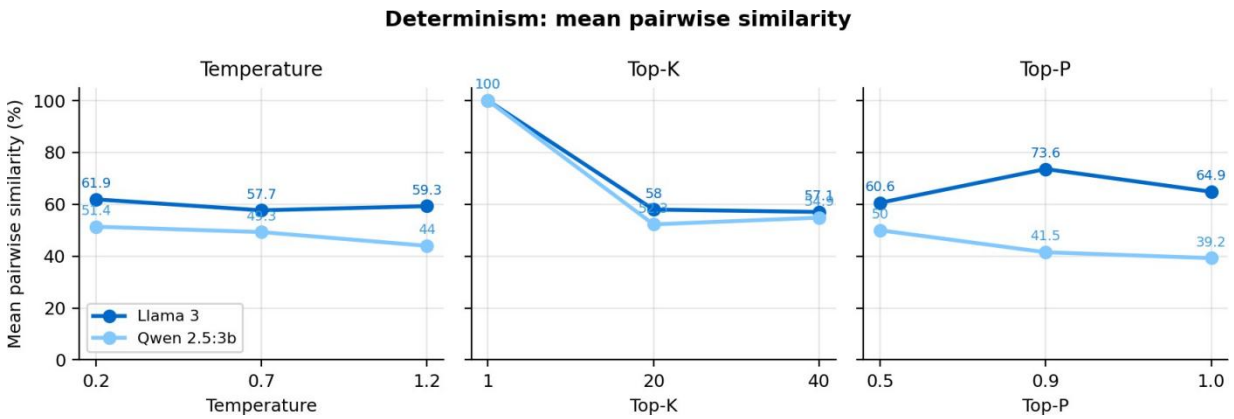


Figure 1. Mean pairwise text similarity (higher means more deterministic).

For Qwen 2.5:3b the expected pattern is visible: similarity falls from 51.4% to 44.0% as Temperature rises from 0.2 to 1.2, and from 50.0% to 39.3% as Top-P rises from 0.5 to 1.0. Llama 3 is more consistent overall (about 57% to 74% outside Top-K = 1) but does not show a clean trend: its Top-P = 0.9 point (73.6%) is higher than both neighbours, and its three baseline runs (57.7%, 57.1% and 73.6%) differ from each other even though the setting is identical. This shows that, with only three runs of one prompt, differences of roughly 10 to 15 points are within noise.

Llama 3 writes longer answers (roughly 180 to 200 words, which is close to the token limit) than Qwen (roughly 95 to 190 words). Because Llama often reaches the length limit, part of its higher similarity may come from shared opening text rather than sampling alone.

Table 1. Determinism summary

Model	Parameter	Setting	Similarity %	Diversity %	Avg words
Llama 3	Temperature	T = 0.2	61.9	38.1	191.3
Llama 3	Temperature	T = 0.7	57.7	42.3	201.3
Llama 3	Temperature	T = 1.2	59.3	40.7	195.3
Llama 3	Top-K	K = 1	100.0	0.0	202.0
Llama 3	Top-K	K = 20	58.0	42.0	187.0
Llama 3	Top-K	K = 40	57.1	42.9	196.3
Llama 3	Top-P	P = 0.5	60.6	39.4	178.7
Llama 3	Top-P	P = 0.9	73.6	26.4	184.7
Llama 3	Top-P	P = 1.0	64.9	35.1	197.3
Qwen 2.5:3b	Temperature	T = 0.2	51.4	48.6	107.0
Qwen 2.5:3b	Temperature	T = 0.7	49.3	50.7	142.3
Qwen 2.5:3b	Temperature	T = 1.2	44.0	56.0	116.3
Qwen 2.5:3b	Top-K	K = 1	100.0	0.0	111.0
Qwen 2.5:3b	Top-K	K = 20	52.3	47.7	129.3
Qwen 2.5:3b	Top-K	K = 40	54.9	45.1	186.7
Qwen 2.5:3b	Top-P	P = 0.5	50.0	50.0	163.0
Qwen 2.5:3b	Top-P	P = 0.9	41.5	58.5	157.0
Qwen 2.5:3b	Top-P	P = 1.0	39.2	60.8	94.3

4.2 Hallucination

Figure 2 shows the hallucination rate for each setting. Llama 3 stays between 0% and 8.3%, while Qwen 2.5:3b stays between 12.5% and 20.8%. Qwen's rate rises steadily with Temperature (12.5%, 16.7%, 20.8%) and jumps at Top-P = 1.0 (20.8%), which matches the expectation that allowing more low-probability tokens increases errors. Top-K had no measurable effect on Qwen (12.5% at every value, including greedy decoding).

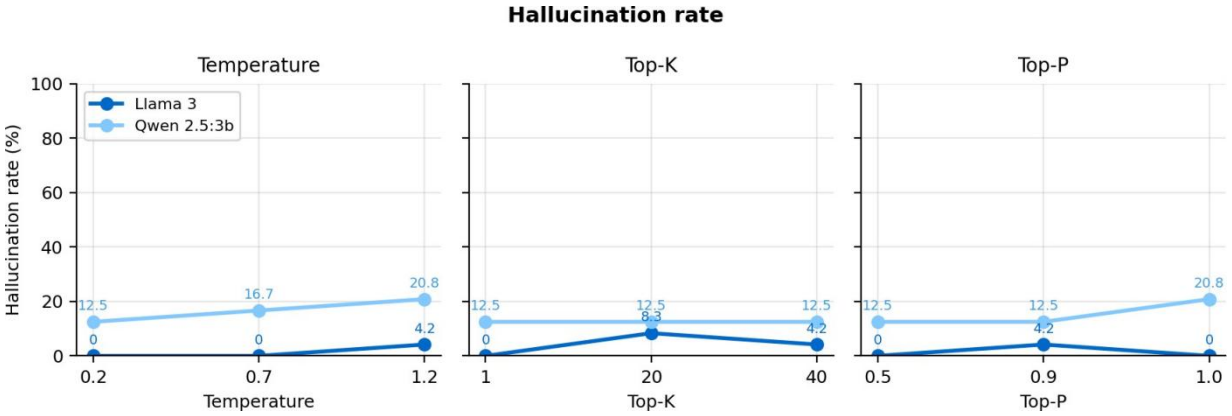


Figure 2. Hallucination rate by setting (lower is better).

The errors are concentrated in a few questions. Both models answered all five factual questions correctly in every run. Qwen invented a summary of the non-existent paper by Dr. Vexley in 100% of its runs (H1), regardless of the setting. This is why Top-K = 1 does not remove its hallucinations: the model's most likely answer is already wrong, so greedy decoding simply repeats the same mistake.

Important note on the metric. The question about the 1987 Nobel Prize in Mathematics (H2) was flagged in 18.5% of runs for both models, but the two cases are very different. All five flagged Llama 3 answers say that the prize was not awarded, for example *The Nobel Prize in Mathematics was not awarded in 1987*. This is a correct answer that the keyword filter does not recognise, so it is a false positive. The five flagged Qwen answers, in contrast, name an invented winner (for example Robert Langlands or Satoshi Kakutani), which are genuine hallucinations. After this manual check Llama 3 produced no genuine hallucinations in 216 answers, while Qwen produced 32 out of 216 (14.8%). The charts above show the unadjusted automatic values, so Llama's small non-zero bars should be read as metric noise rather than model errors.

Table 2. Hallucination summary (automatic metric)

Model	Parameter	Setting	Accuracy %	Halluc. %	Avg words
Llama 3	Temperature	T = 0.2	100.0	0.0	14.8
Llama 3	Temperature	T = 0.7	100.0	0.0	16.6
Llama 3	Temperature	T = 1.2	95.8	4.2	16.0
Llama 3	Top-K	K = 1	100.0	0.0	13.6
Llama 3	Top-K	K = 20	91.7	8.3	16.3
Llama 3	Top-K	K = 40	95.8	4.2	16.6
Llama 3	Top-P	P = 0.5	100.0	0.0	13.6
Llama 3	Top-P	P = 0.9	95.8	4.2	15.3
Llama 3	Top-P	P = 1.0	100.0	0.0	16.4
Qwen 2.5:3b	Temperature	T = 0.2	87.5	12.5	20.8
Qwen 2.5:3b	Temperature	T = 0.7	83.3	16.7	19.3
Qwen 2.5:3b	Temperature	T = 1.2	79.2	20.8	17.1
Qwen 2.5:3b	Top-K	K = 1	87.5	12.5	19.6
Qwen 2.5:3b	Top-K	K = 20	87.5	12.5	18.7
Qwen 2.5:3b	Top-K	K = 40	87.5	12.5	19.5
Qwen 2.5:3b	Top-P	P = 0.5	87.5	12.5	19.1
Qwen 2.5:3b	Top-P	P = 0.9	87.5	12.5	20.5
Qwen 2.5:3b	Top-P	P = 1.0	79.2	20.8	18.6

4.3 Structure

Qwen 2.5:3b produced a valid strict structure in all 54 answers (100%), in every setting. Llama 3 passed all 27 numbered-list tasks but only 21 of 27 JSON tasks (77.8%). In the strict JSON task, five of the six failures were answers that ended with the closing bracket of the examples list but were missing the final closing brace, so the object was not valid JSON; the sixth was cut off in the middle of a string at Temperature 1.2, probably because it reached the token limit.

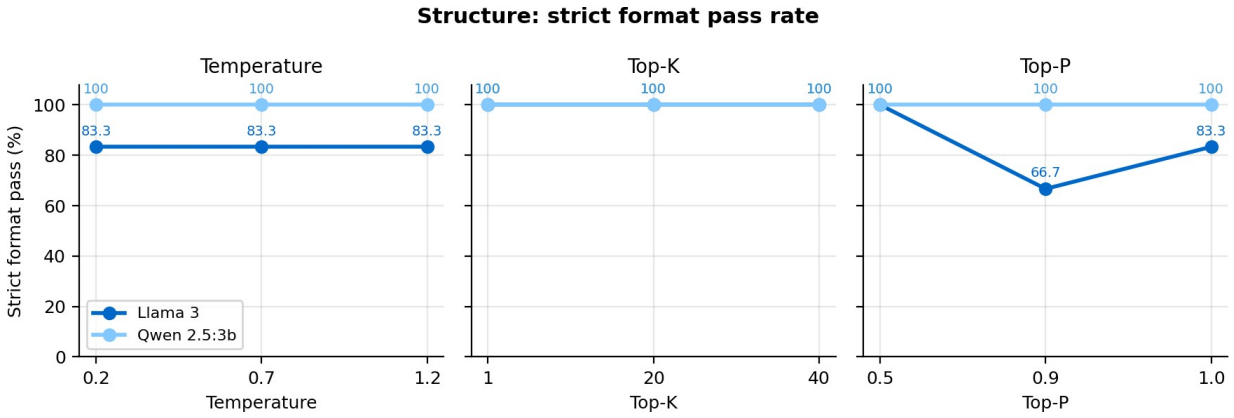


Figure 3. Strict format pass rate (higher is better).

Llama's failures are spread over Temperature (83.3% at all three values), Top-P 0.9 (66.7%) and Top-P 1.0 (83.3%), while it passed every task at Top-K 1, 20 and 40 and at Top-P 0.5. The Temperature sweep shows the same pass rate at all values, so the data do not support a clear link between Temperature and format errors. The lower pass rate at higher Top-P is consistent with the expectation, but with six tasks per setting a single failure changes the rate by 16.7 points, so this should be treated as weak evidence.

Table 3. Structure summary

Model	Parameter	Setting	Parseable %	Strict pass %	Avg words
Llama 3	Temperature	T = 0.2	83.3	83.3	86.5
Llama 3	Temperature	T = 0.7	83.3	83.3	81.3
Llama 3	Temperature	T = 1.2	83.3	83.3	104.7
Llama 3	Top-K	K = 1	100.0	100.0	87.0
Llama 3	Top-K	K = 20	100.0	100.0	78.5
Llama 3	Top-K	K = 40	100.0	100.0	90.8
Llama 3	Top-P	P = 0.5	100.0	100.0	83.2
Llama 3	Top-P	P = 0.9	66.7	66.7	75.7
Llama 3	Top-P	P = 1.0	83.3	83.3	76.2
Qwen 2.5:3b	Temperature	T = 0.2	100.0	100.0	70.8
Qwen 2.5:3b	Temperature	T = 0.7	100.0	100.0	60.8
Qwen 2.5:3b	Temperature	T = 1.2	100.0	100.0	61.5
Qwen 2.5:3b	Top-K	K = 1	100.0	100.0	89.0
Qwen 2.5:3b	Top-K	K = 20	100.0	100.0	55.8
Qwen 2.5:3b	Top-K	K = 40	100.0	100.0	52.7
Qwen 2.5:3b	Top-P	P = 0.5	100.0	100.0	57.2
Qwen 2.5:3b	Top-P	P = 0.9	100.0	100.0	55.8
Qwen 2.5:3b	Top-P	P = 1.0	100.0	100.0	61.7

4.4 Overall effect of each parameter

Table 4 compares the three parameters directly. For each metric the value at the lowest setting is subtracted from the value at the highest setting, averaged over both models.

Table 4. Effect sizes (average of Llama 3 and Qwen 2.5:3b)

Metric	Parameter	Low value	High value	Score at low	Score at high	Change (pts)
Determinism (similarity)	Temperature	0.2	1.2	56.7	51.6	-5.0
Determinism (similarity)	Top-K	1	40	100.0	56.0	-44.0
Determinism (similarity)	Top-P	0.5	1	55.3	52.1	-3.2
Hallucination rate	Temperature	0.2	1.2	6.2	12.5	+6.2
Hallucination rate	Top-K	1	40	6.2	8.3	+2.1
Hallucination rate	Top-P	0.5	1	6.2	10.4	+4.2
Structure pass	Temperature	0.2	1.2	91.7	91.7	+0.0
Structure pass	Top-K	1	40	100.0	100.0	+0.0
Structure pass	Top-P	0.5	1	100.0	91.7	-8.3

5. Discussion

Top-K is the strongest lever for determinism. Moving Top-K from 40 down to 1 changed similarity by 44 points on average, an order of magnitude more than Temperature (5 points) or Top-P (3 points). Within these ranges, Top-K = 1 is the only reliable way to obtain reproducible output. Low Temperature alone does not guarantee repeatability while Top-K and Top-P still allow several candidates.

Hallucination depends on the model more than on the sampler. The difference between the two models (about 15 points) is larger than any change caused by the parameters (2 to 6 points on average). Qwen 2.5:3b fails on a specific kind of question, one about a plausible but non-existent paper, and decoding changes cannot fix a wrong most-likely answer. Lowering Temperature and Top-P still gave Qwen a modest benefit (12.5% instead of 20.8% at the extremes), so conservative settings are sensible for factual use.

Format compliance depends on the model and on small details. Qwen 2.5:3b followed both formats perfectly, while Llama 3 tended to forget the closing brace of the JSON object. A practical remedy is to validate and retry, or to use constrained or JSON-mode decoding for structured output.

Practical recommendations.

- Factual question answering and structured output: use a low Temperature (0.2 or lower), a restricted Top-P (about 0.5) and, if repeatability matters, Top-K = 1.
- Creative or varied writing: keep Temperature around 0.7 to 1.0 with Top-K 40 and Top-P 0.9, accepting that outputs will differ between runs.
- Always validate model output (parse JSON, check lists) and never trust a small model on questions about obscure or possibly non-existent facts.

6. Limitations

- **Small sample.** Determinism uses a single prompt and three runs; hallucination has 8 questions and structure has 2 tasks, each repeated three times. Differences of one or two answers move the percentages by several points.
- **Visible run-to-run noise.** The baseline setting (T 0.7, K 40, P 0.9) was run in every sweep and gave different results each time (for example Llama 3 determinism similarity of 57.7%, 57.1% and 73.6%).
- **Keyword-based hallucination metric.** It misses correct answers phrased unusually (such as *was not awarded*) and cannot judge partial correctness, so it should be checked manually.
- **TF-IDF similarity** measures word overlap, not meaning, and the answer length limit can affect it.
- **One-at-a-time sweeps** do not capture interactions between parameters, and only two small models were tested.

7. Conclusion

The experiments confirm that Top-K = 1 gives fully deterministic output and that the smaller Qwen 2.5:3b model hallucinates substantially more than Llama 3, particularly on questions about non-existent entities, with a mild increase as Temperature and Top-P rise. Strict format compliance was high for both models, with Llama 3 showing occasional JSON errors. The effects of Temperature and Top-P on most metrics were small relative to sampling noise, so larger experiments (more prompts and more runs) would be needed to confirm them. For reliable results the choice of model and the validation of outputs matter at least as much as the exact sampling parameters.